

HATEFUL MEME DETECTION

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ABSTRACT. Hateful Meme Detection, detecting hate speech in multimodal memes is a hard but essential task in which we combine visual and linguistic features in order to gauge the intrinsic meaning of the meme. Using the Hateful Meme Challenge dataset by Meta AI, we have experimented with various multimodal fusion techniques such as early fusion, mid fusion, and late fusion and compared their performance with unimodal models which exclusively use either images or text. We have used pre-trained RoBERTa to generate sentence embeddings, and pre-trained ResNext to generate image embeddings. As an improvement, we have used CLIP embeddings from images and text which gives better results. We use accuracy and AUCROC as our metric and reach best accuracy of 66.8% and AUCROC of 0.78 respectively on the test data. Early fusion gives the best results, followed by mid fusion and then late fusion. All the multimodal techniques beat the unimodal models which makes it evident that the task requires both the modalities - vision and language.

1. INTRODUCTION

The rise of social media, has led to the invention a new form of entertainment through image media. Memes covers a vast range of topic, in a quirky way, and it can range from little light-hearted jokes, to sometimes staying on the edge of dark humour, they may lead into darker realms, where they are used by trolls to make fun of a person/community on anything and everything, spreading hate towards an individual or a community. We attempt to detect such hateful memes by trying various models and architectural techniques. Most memes are a combination of an image with some text on it, and the image and text combined make the meme meaningful. This becomes a hard task for a machine to detect whether it is hateful or not. For example:



FIGURE 1. Benign confounder memes

In the above examples, the combination of text and image conveys different meaning as compared to either of them individually. Therefore, it is essential to use a multimodal model that uses both image and text modalities to detect hateful memes. We have attempted multiple multimodal architectures - early fusion, mid fusion, and late fusion and compared them to using unimodal models which are explained in detail in the Approach section of the report.

Hateful Memes challenge hosted by Meta AI provides us with a dataset consisting of various meme images along with their text which we have used to build our model.

1.1. Contributions. We experimented with various unimodal and multimodal architectures like early-fusion, mid-fusion and late-fusion and concluded that all the multimodal architectures out performs unimodal ones. Among the multimodal architectures early fusion performs the best. All of these experiments were run first using RoBERTa and ResNeXt embeddings and then using CLIP embeddings. Early fusion with CLIP embeddings out performed all the remaining architectures and gave an accuracy score of 66.8% and an AUROC of 0.78 on the Hateful Memes challenge dataset.

2. BACKGROUND

Facebook released a challenge for detecting hateful multimodal memes in 2020 - Facebook Blog for the challenge to detect hate speech on its platform and make it a safer place. A meme is usually an image, followed with some text or caption which brings more context to the image, and often doesn't really negate any information without text. While memes can be difficult to decode, since lot of them have unknown context hidden to references only known in 'pop-culture'. A common internet slang, which is used to describe the memes which are popular, and edgy, or in the **grey area** is **dank**.

Using robust image and text networks like ResNext and RoBERTa, we would try to find if we can still decode the images and text, and then together with them, find a way to correctly classify them.

3. RELATED WORK

Previously, lots of work has been done on hate speech detection in images and text separately. Our project was to detect hate speech in an image that had text embedded inside it. We experimented using some of the multimodal architectures using pretrained embeddings for image and text.

Another similar work is done in caption generator[4], for which one of the most common dataset is **Flickr-8**. We also tried using the caption generating model using Flickr-8, and passed our images, but with the complexity and the artistic implementation of the design in memes, the captions were way far from describing the elements in our data.

Here are the results released by Facebook on its hateful memes dataset: As mentioned by the winners of Facebook

TYPE	MODEL	TEST Acc.	AUROC
	Human	84.70	82.65
Unimodal	Image-grid	52.00	52.63
	Image-region	52.13	55.92
	Text BERT	59.20	65.08
Multimodal (Unimodal pretraining)	Late fusion	59.66	64.75
	Concat BERT	59.13	65.79
	MMBT-grid	60.06	67.92
	MMBT-region	60.23	70.73
	VILBERT	62.30	70.45
	Visual BERT	63.20	71.33
Multimodal (Multimodal pretraining)	VILBERT CC	61.10	70.03
	Visual BERT COCO	64.73	71.41

FIGURE 2. Facebook hate meme challenge result

hate memes challenge they used ensembles of state-of-the-art vision and language models such as VILLA, UNITER, ERNIE-ViL and VL-BERT works the best for this problem but due to limited resources we could not use those.

We only used embeddings of pretrained ResNext-Roberta and CLIP model and fine tuned some multimodal architectures as mentioned in the Approach and tried to get closer to their results.

We also tried generating caption [6] from the image and using it along with the meme embedded text but this did not give good result probably because the weightage given to images increased significantly and the dataset was designed such that it will not do well on unimodal models.

4. APPROACH

Using the Hateful Meme Challenge dataset by Meta AI, we have experimented with multimodal and various multimodal fusion techniques such as early fusion, mid fusion, and late fusion.

Our general approach was to get embeddings of images and text using popular pre-trained models, which are basically a vector in a vector space representing various features of our data. Hence, we have used ResNext to generate image embeddings - this takes the image as input and generates an embedding of the image of size 2048. For text, we have used RoBERTa to give us sentence embeddings - this takes the text (sentence or phrase) as input and generates an embedding of size 768.

Moreover, we have also used CLIP to generate image and text embeddings as it specializes in multimodal tasks of image and text. CLIP gives us embeddings of image and text of size 512 each.

These embeddings fused in various ways form the input to our classifier.

Our approach does not involve fine-tuning any pre-trained models for this task. Rather, we focus on the power of a multi-layer perceptron using these embeddings. We want to check if simple neural networks can be as powerful as fine-tuning huge and strong pretrained models. So, throughout our models, we have used multi-layer perceptrons, with ReLU non-linearities and BatchNorms.

4.1. Unimodal Models: In this case we train two separate models for text and image and see how they perform individually.

4.1.1. Text unimodal: We extracted the image embedded text embeddings using pretrained RoBERTa. This was given as input to a 3 layer MLP with relu and sigmoid as activations. The output predictions are compared with the results. Same procedure using pretrained CLIP embeddings instead of RoBERTa.

4.1.2. Image unimodal: The exact same procedure as unimodal text model was followed just the input Roberta and CLIP embeddings were replaced with pretrained ResNext and CLIP image embeddings.

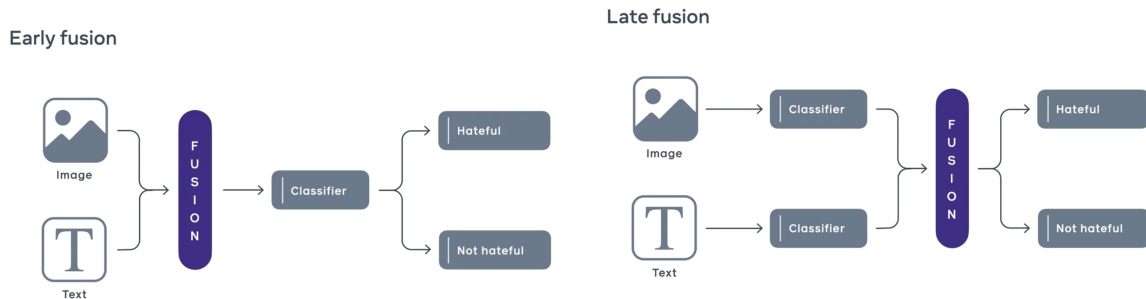
4.2. Multimodal Models: We have experimented with multimodal architectures to detect hateful memes - multimodal models like early fusion, late fusion and mid fusion. As the name suggests, multimodal models fuse multiple modalities to perform a task. In our case, the different modals are images and text. There are three main ways to fuse the different modals based on when you fuse them together - early fusion, mid fusion, and late fusion.[1] These approaches will be defined below in more detail.

4.2.1. Early Fusion: Initially, we moved to train a classifier that would intake 2047 embedding from the images generated by ResNext, and 768, by Roberta. We concatenated them as an array and passed them to a multi-layer classifier, using ReLU as activation layer. We used *Pytorch* for all the models, and trained it on both Adam and SGD optimizers.

To generate better results we switched to textbfCLIP by Open AI. as it generates embeddings for both text and images by itself, and also look for co-relations.

4.2.2. Mid Fusion: In this method, first the neural network was such that, it took the text and image embeddings as input. They were passed through two separate Linear layers with relu activation and the outputs are two 64 dimensional vectors for text and image separately. Now these two vectors were concatenated and further passed through the a linear layer to generate an output with sigmoid activation. So in this approach only one neural network is trained and the image and text embeddings are getting concatenated somewhere in between.

4.2.3. Late Fusion: In late fusion, we basically trained two MLP models separately for text and image embeddings. The MLP models consisted of 3 linear layers and relu as their activation functions. The output of each MLP was a single dimantional vector with sigmoid as activation. The two output scores were averaged during inference to get a prediction.



5. EXPERIMENTAL RESULTS

We used the facebook Hateful Memes Challenge dataset for this project. The dataset had multimodal memes and facebook had designed this dataset specifically such that unimodal models do not perform well separately. We experimented with various multimodal fusion techniques such as early fusion, mid fusion, and late fusion and compared their performance with unimodal models which exclusively use either images or text. All the below mentioned models

were trained using different hyperparameters and here are the best results we obtained using adams optimizer and a learning rate of $1e-4$:

Architecture	Epochs	Val Acc.	Val AUCROC	Test Acc.	Test AUCROC
Text unimodal	100	56.60%	0.61	59.10%	0.63
Image unimodal	50	52.80%	0.54	52.90%	0.54
Early fusion	100	61.00%	0.72	63.90%	0.73
Mid fusion	100	59.20%	0.63	61.30%	0.64
Late fusion	100	58.20%	0.62	59.50%	0.63

TABLE 1. Results using RoBERTa and ResNeXt embeddings

Architecture	Epochs	Val Acc.	Val AUROC	Test Acc.	Test AUROC
Text unimodal	100	57.40%	0.61	58.40%	0.63
Image unimodal	100	59.20%	0.62	60.20%	0.64
Early fusion	100	64.20%	0.74	66.80%	0.78
Mid fusion	170	63.20%	0.67	64.80%	0.71
Late fusion	170	62.60%	0.66	63.00%	0.69

TABLE 2. Results using CLIP embeddings

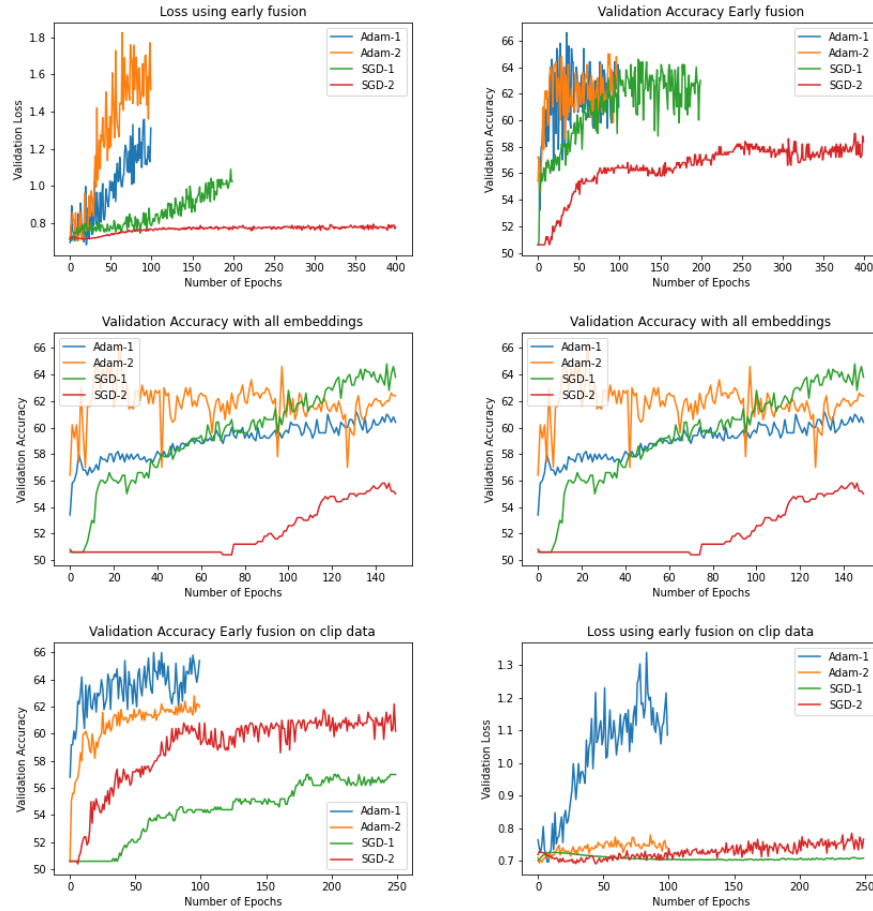
As it is visible from the results that early fusion performed better than all other models. It outperformed unimodal simply because they did not take both text and linguistic reference into consideration. The late fusion model could not perform that well is probably because the embeddings were trained separately so in case of images like the benign confounder shown above it could not predict correct answers.

Now to further improve the results we decided to dive deeper into the early fusion model and tried various experiments. Here are validation loss from various models we used. In case of early fusion, we trained using the embeddings from ResNext, and RoBerta. In this, we have the best of each optimizer for all cases as follows:

Embeddings Used	Optimizer	Epochs	Val Acc.	Val AUCROC	Test Acc.	Test AUCROC
ResNext and RoBerta	Adam	100	61%	0.72	63.9%	0.73
ResNext and RoBerta	SGD	200	63.00%	0.70	64.5%	0.72
ResNext,RoBerta,Clip	Adam	150	58.40%	0.66	63.50%	0.73
ResNext,RoBerta,Clip	SGD	150	59.60%	0.69	63.80%	0.72
Only Clip	Adam	100	64.2%	0.74	66.8%	0.78
Only Clip	SGD	250	60.2%	0.69	62.2%	0.73

TABLE 3. Results for all Early Fusion models

As we can see, using only Clip gives us the best result with Adam Optimizer. The increase in validation loss is due to over-fitting. Therefore, one thing which we could understand is that these embeddings are not enough for us to co-relate or understand the complexity of the data, or there are other ways which would have better relationship between data form the two given sources. Here are the graphs we obtained from validation loss and accuracy while training all the Early fusion models with different optimizers as per the order in the table:



Our data is overfitting due to use of over thousand feature vectors. We tried using dropout to reduce it to an extent, but that didn't change much about our final accuracy. Due to this we can still notice the loss increasing over time. It is possibly because while using the multimodal model, the assigning of weights is not done very well. There could be number of reasons for that. We can explain the validation loss increment with over-fitting over other modals. Although while using Clip, we didn't face that problem much, making it very clear that clip represents the features in the best way to solve our classification problem. Clip is also better because it has smaller space which makes it faster, and has less chances of over-fitting. It implements a masked self-attention transformer for text-encoding, which doesn't allow text to be iterated multiple times while encoding the images, giving a smaller feature vector space, making our particular implementation better.

6. DISCUSSION

We experimented with both unimodal and multimodal approaches which makes it clear to us that the task of Hateful Memes is much better solved using both image and textual features. CLIP, trained on natural language data connecting images to text, performs much better on the test set as both the features are from the same distribution as compared to getting image features from ResNeXt and text features from RoBERTa. However, we use the same type of linear layers for both the approaches and increasing the number of linear layers overfits the model because of the small datasize. To achieve better results, given more time, we wish to either play with other architecture. We could implement various dropouts and implemented more layers to create a filter of relation. Or we could train one of the existing model by itself instead of generating embedding, and build a classifier as the final layer. We could also look for more meme related database to train our network to have better understanding of the memes itself, to generated better embeddings.

REFERENCES

- (1) Facebook Blog for the challenge

- (2) Douwe Kiela , Hamed Firoozravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, Davide Testuggine: The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes. arxiv:2005.04790
- (3) Tanvi Deshpande, Nitya Mani: An Interpretable Approach to Hateful Meme Detection. arxiv:2108.10069
- (4) MIT 6.S191: Multimodal Deep Learning Lecture
- (5) The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes D. Kiela, H. Firooz, A. Mohan, V. Goswami, A. Singh, P. Ringshia, D. Testuggine
- (6) Generating images from caption and vice versa